

Prabhat Kumar

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LinkedIn • Github • Leetcode



PROFESSIONAL SUMMARY

B.Tech student with skills in AI/ML and full stack development. Experience in building web applications and machine learning projects using React, Next.js, and Python. Strong understanding of data structures and problem solving. Focused on learning, improving, and building real-world solutions.

EDUCATION

Bachelor of Technology (Artificial intelligence)
Newton School of Technology, Rishihood University

2023 - 2027
Grade: 7.0/10.0

INTERNSHIPS

AI & ML INTERN
Indian School of business

March 2025 - March 2026
Gachibowli, Hyderabad

- **Developed NLP-based judgment analysis system** using transformer models (Legal-BERT) to classify case outcomes and extract insights
- **Built loan defaulter prediction model** using ML algorithms (Random Forest/XGBoost) on financial datasets
- **Automated data scraping using Selenium** to collect large-scale legal judgments and financial data with reliable workflows.
- **Designed end-to-end ML pipelines** covering preprocessing, feature engineering, training, evaluation, and inference.
- **Optimized model performance** using hyperparameter tuning, class imbalance handling (SMOTE), and metrics like F1-score & ROC-AUC

PROJECTS

Vertical Eden Garden ([Github](#)) ([Demo](#))

April 2026

- **Full-Stack MERN Platform** enabling users to browse gardening services, book appointments, and make secure online payments seamlessly.
- **End-to-End Customer Workflow** with service discovery, 3-step booking system, real-time tracking, invoice generation, and review system.
- **Robust Authentication System** supporting Email/Password and OTP-based login with secure JWT-based authorization.
- **Admin Dashboard Control Panel** for managing users, bookings, and revenue with role-based access and analytics insights.
- **Integrated Smart Features** including AI chatbot (Gemini), Razorpay payments, WhatsApp integration, and responsive modern UI.

Memory Visualisation Allocators ([Github](#)) ([Demo](#))

March 2026

- Built an interactive **OS memory allocation simulator** using Next.js and TypeScript
- Implemented **First Fit, Best Fit, and Worst Fit** allocation algorithms with dynamic process handling
- Designed a **custom simulation engine** for memory allocation, deallocation, compaction, and lifecycle management
- Developed a **scalable, component-based architecture** with optimized state management and real-time visualization

Nexus ([Github](#)) ([Demo](#))

April 2024

- Architected and developed a **full-stack real-time social media platform** using React, Node.js, PostgreSQL, and Socket.IO, handling end-to-end system design and implementation.
- Engineered **secure authentication and authorization flow** with JWT (access + refresh tokens), bcrypt hashing, and protected routes to ensure production-level security.
- Designed and implemented **scalable RESTful APIs and database models** using Express.js and Sequelize ORM, supporting complete CRUD operations for users, posts, messages, and notifications.
- Built **real-time features (chat, notifications, online status)** using Socket.IO, ensuring low-latency communication and seamless user experience.
- Led **UI/UX development with responsive design and performance optimization**, including infinite scrolling, debounce search, and modern Tailwind-based inter

SKILLS

AI / ML: Machine Learning

APIs & Testing: Postman, Playwright, API

Cloud & DevOps: Git and Github

Computer Languages: TypeScript, HTML, CSS, JavaScript, Python, C, Java, SQL

Courses: Data Structure

Databases & Data Tools: Firebase, NoSQL, Spreadsheet, Excel, MySQL, MongoDB

Domain & Business Skills: Figma

Frameworks & Libraries: Pandas, Matplotlib, Tailwind, Next.js, Tailwind CSS, OpenCV, Next JS, Prisma ORM, NumPy, React, Node.js, Express JS

Soft Skills: Leadership, Decision-making, Responsibility, Presentation Skills

EXTRA-CURRICULAR ACTIVITIES

- **Planned and executed Republic Day event**, coordinating with alumni and industry professionals to ensure smooth operations and high engagement
- **Facilitated professional networking opportunities** by connecting students, alumni, and industry experts, encouraging collaboration and career growth
- **Managed event logistics and coordination**, including communication, scheduling, and stakeholder handling
- **Strengthened community engagement** by increasing participation and fostering long-term alumni-student connections